

Prazosin

## Therapeutics

### Brands

- Minipress
- *see index for additional brand names*

### Generic?

Yes

### Class

- Alpha 1 adrenergic blocker

### Commonly Prescribed for

*(bold for FDA approved)*

- **Hypertension**
- Nightmares associated with PTSD
- Blood circulation disorders
- Problems urinating due to enlarged prostate
- Passing of kidney stones



### How the Drug Works

- Blocks alpha 1 adrenergic receptors to reduce noradrenergic hyperactivation
- Stimulation of central noradrenergic receptors during sleep may activate traumatic memories, so blocking this activation may reduce nightmares

### **How Long Until It Works**

- Within a few days to a few weeks

### **If It Works**

- Reduces the severity and frequency of nightmares associated with PTSD

### **If It Doesn't Work**

- Increase dose
- Switch to another agent



### **Best Augmenting Combos for Partial Response or Treatment Resistance**

- Prazosin is itself an adjunct agent for the treatment of nightmares associated with PTSD

### **Tests**

- None for healthy individuals

- False-positive results may occur in screening tests for pheochromocytoma in patients who are being treated with prazosin; if an elevated urinary VMA is found, prazosin should be discontinued and the patient retested after a month

## Side Effects

### How Drug Causes Side Effects

- Excessive blockade of alpha 1 peripheral noradrenergic receptors

### Notable Side Effects

- Dizziness, lightheadedness, headache, fatigue, blurred vision
- Nausea



### Life-Threatening or Dangerous Side Effects

- Syncope with sudden loss of consciousness

### Weight Gain



- Reported but not expected

### Sedation



- Occurs in significant minority

## **What to Do About Side Effects**

- Lower the dose
- Wait
- Wait
- Wait
- In a few weeks, switch to another agent

## **Best Augmenting Agents for Side Effects**

- Often best to try another treatment prior to resorting to augmentation strategies to treat side effects

## **Dosing and Use**

### **Usual Dosage Range**

- 1–16 mg/day, generally in divided doses

### **Dosage Forms**

- Capsule 1 mg, 2 mg, 5 mg

### **How to Dose**

- Formal dosing recommendations for treating nightmares have not been established

- Initial 1 mg at bedtime; increase dose (divided) until the nightmares resolve or an intolerable side effect occurs



### **Dosing Tips**

- Dosing may be extremely individualized, with as little as 2 mg/day helpful for some patients and as much as 40 mg/day necessary for other patients
- Therapeutic dose does not correlate with blood levels
- Divided dosing may be preferable; in particular, giving a smaller dose during the day may be beneficial if the patient has persistent hyperarousal and re-experiencing symptoms during the day
- Risk of syncope can be decreased by limiting the initial dose to 1 mg and using slow dose titration

### **Overdose**

- No fatalities have been reported; sedation, depressive reflexes, hypotension

### **Long-Term Use**

- Has not been evaluated in controlled studies
- Nightmares may return if prazosin is stopped

### **Habit Forming**

- No

## **How to Stop**

- Taper to avoid hypertension

## **Pharmacokinetics**

- Elimination half-life 2–3 hours



## **Drug Interactions**

- Concomitant use with a phosphodiesterase-5 (PDE-5) inhibitor can have additive effects on blood pressure, potentially leading to hypotension; thus, a PDE-5 inhibitor should be initiated at the lowest possible dose
- Concomitant use with a beta blocker (e.g., propranolol) can have additive effects on blood pressure
- Concomitant use with other alpha 1 blockers, which include many psychotropic agents, can have additive effects leading to hypotension



## **Other Warnings/Precautions**

- Prazosin can cause syncope with sudden loss of consciousness, most often in association with rapid dose increases or the introduction of another antihypertensive drug
- Intraoperative floppy iris syndrome (IFIS) has been observed during cataract surgery in some patients treated with alpha 1 adrenergic blockers, which may require modifications to the surgical technique;

however, there does not appear to be a benefit of stopping the alpha 1 adrenergic blocker prior to cataract surgery

- Avoid situations that can cause orthostatic hypotension, such as extensive periods of standing, prolonged or intense exercise, and exposure to heat

### **Do Not Use**

- If there is a proven allergy to quinazolines or prazosin

## **Special Populations**

### **Renal Impairment**

- Use with caution in patients with severe impairment
- May require lower dose

### **Hepatic Impairment**

- Use with caution

### **Cardiac Impairment**

- Use with caution in patients who are predisposed to hypotensive or syncopal episodes

### **Elderly**

- Some patients may tolerate lower doses better

- Higher risk of orthostatic hypotension and syncope



### **Children and Adolescents**

- Safety and efficacy have not been established



### **Pregnancy**

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Prazosin has been used alone or in combination with other hypotensive agents in severe hypertension of pregnancy, with no fetal or neonatal abnormalities reported
- Prazosin should be used during pregnancy only if the potential benefits justify the potential risks to the mother and fetus

### **Breast Feeding**

- Some drug is present in breast milk
- If child becomes irritable or sedated, breast feeding or drug may need to be discontinued



- Must weigh benefits of breast feeding with risks and benefits of treatment versus nontreatment to both the infant and the mother

## **The Art of Psychopharmacology**

### **Potential Advantages**

- For patients with PTSD who do not respond to SSRIs/SNRIs or exposure therapy
- Specifically for nightmares and other symptoms of autonomic arousal

### **Potential Disadvantages**

- Patients with cardiovascular disease
- Patients taking concomitant psychotropic drugs with alpha 1 antagonist properties

### **Primary Target Symptoms**

- Nightmares



### **Pearls**

- The evidence base for using prazosin to treat nightmares associated with PTSD is limited but positive, and prazosin is recommended by the Department of Veterans Affairs as an adjunct treatment for this purpose

- Initiate treatment early in the onset of nightmares following exposure to trauma
- May also be useful for nightmares and symptoms of autonomic arousal in other trauma and stress-related disorders in addition to PTSD

### *Suggested Reading*

Kung S , Espinel Z , Lapid MI . Treatment of nightmares with prazosin: a systematic review . *Mayo Clin Proc* 2012 ;**87** (9 ):890 –900 .

Schoenfeld FB , Deviva JC , Manber R . Treatment of sleep disturbances in posttraumatic stress disorder: a review . *J Rehabil Res Dev* 2012 ;**49** (5 ):729–52 .

Van Berkel VM , Bevelander SE , Mommersteeg PM . Placebo-controlled comparison of prazosin and cognitive-behavioral treatments for sleep disturbances in US Military Veterans . *J Psychosom Res* 2012 ;**73** (2 ):153 .

Pregabalin

## Therapeutics

### Brands

- Lyrica
- Lyrica CR
- *see index for additional brand names*

### Generic?

Yes

### Class

- Neuroscience-based Nomenclature: glutamate voltage-gated calcium channel blocker (Glu-CB)
- Anticonvulsant, antineuralgic for chronic pain, alpha 2 delta ligand at voltage-sensitive calcium channels

### Commonly Prescribed for

*(bold for FDA approved)*

- **Diabetic peripheral neuropathy (IR and CR)**
- **Postherpetic neuralgia (IR and CR)**
- **Fibromyalgia (IR)**

- **Neuropathic pain associated with spinal cord injury (IR)**
- **Partial onset seizures in adults and pediatric patients ages 1 month and older (IR, adjunctive)**
- Peripheral neuropathic pain
- Generalized anxiety disorder (GAD)
- Panic disorder
- Social anxiety disorder

### **How the Drug Works**

- Pregabalin is a leucine analog and is transported both into the blood from the gut and also across the blood–brain barrier into the brain by the system L transport system (a sodium-independent transporter) as well as by additional sodium-dependent amino acid transporter systems
- \* Binds to the alpha 2 delta subunit of voltage-sensitive calcium channels
- This closes N and P/Q presynaptic calcium channels, diminishing excessive neuronal activity and neurotransmitter release
- Although structurally related to gamma-aminobutyric acid (GABA), no known direct actions on GABA or its receptors

### **How Long Until It Works**

- Can reduce neuropathic pain and anxiety within a week

- Should reduce seizures by 2 weeks
- If it is not producing clinical benefits within 6–8 weeks, it may require a dosage increase or it may not work at all

### **If It Works**

- The goal of treatment of neuropathic pain, seizures, and anxiety disorders is to reduce symptoms as much as possible, and if necessary in combination with other treatments
- Treatment of neuropathic pain most often reduces but does not eliminate all symptoms and is not a cure since symptoms usually recur after medicine stopped
- Continue treatment until all symptoms are gone or until improvement is stable and then continue treating indefinitely as long as improvement persists

### **If It Doesn't Work (for Neuropathic Pain)**

- Many patients have only a partial response where some symptoms are improved but others persist
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider biofeedback or hypnosis for pain
- Consider psychotherapy for anxiety

- Consider the presence of noncompliance and counsel patient
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)



### **Best Augmenting Combos for Partial Response or Treatment Resistance**

- \* In addition to being a first-line treatment for neuropathic pain and anxiety disorders, pregabalin is itself an augmenting agent to numerous other anticonvulsants in treating epilepsy
- For postherpetic neuralgia, pregabalin can decrease concomitant opiate use
- \* For neuropathic pain, TCAs and SNRIs as well as tiagabine, other anticonvulsants, and even opiates can augment pregabalin if done by experts while carefully monitoring in difficult cases
- For anxiety, SSRIs, SNRIs, or benzodiazepines can augment pregabalin

### **Tests**

- None for healthy individuals

## **Side Effects**

### **How Drug Causes Side Effects**

- CNS side effects may be due to excessive blockade of voltage-sensitive calcium channels

### **Notable Side Effects**

#### **\* Sedation, dizziness**

- Ataxia, fatigue, tremor, dysarthria, paresthesia, memory impairment, coordination abnormal, impaired attention, confusion, euphoric mood, irritability
- Vomiting, dry mouth, constipation, weight gain, increased appetite, flatulence
- Blurred vision, diplopia
- Peripheral edema
- Libido decreased, erectile dysfunction



### **Life-Threatening or Dangerous Side Effects**

- Rare activation of suicidal ideation and behavior (suicidality)

### **Weight Gain**



- Occurs in significant minority

### **Sedation**



- Many experience and/or can be significant in amount

- Dose-related
- Can wear off with time

### **What to Do About Side Effects**

- Wait
- Wait
- Wait
- Take more of the dose at night to reduce daytime sedation
- Lower the dose
- Switch to another agent

### **Best Augmenting Agents for Side Effects**

- Many side effects cannot be improved with an augmenting agent

## **Dosing and Use**

### **Usual Dosage Range**

- IR: 150–600 mg/day in 2–3 doses
- CR: 330 mg once per day

### **Dosage Forms**

- Capsule 25 mg, 50 mg, 75 mg, 100 mg, 150 mg, 200 mg, 225 mg, 300 mg



- Oral solution 20 mg/mL
- Extended-release tablet 82.5 mg, 165 mg, 330 mg

### **How to Dose**

- Neuropathic pain (IR): initial 150 mg/day in 2–3 doses; can increase to 300 mg/day in 2–3 doses after 7 days; can increase to 600 mg/day in 2–3 doses after 7 more days; maximum dose generally 600 mg/day (may be lower for diabetic peripheral neuropathy and fibromyalgia)
- Neuropathic pain (CR): 165 mg once per day; can increase to 330 mg once per day within a week; maximum dose 660 mg once per day for postherpetic neuralgia
- Seizures (adults): initial 150 mg/day in 2–3 doses; can increase to 300 mg/day in 2–3 doses after 7 days; can increase to 600 mg/day in 2–3 doses after 7 more days; maximum dose generally 600 mg/day

### **Dosing Tips**

- \* Generally given in one-third to one-sixth the dose of gabapentin
- If pregabalin is added to a second sedating agent, such as another anticonvulsant, a benzodiazepine, or an opiate, the titration period should be at least a week to improve tolerance to sedation
- Most patients need to take pregabalin IR only twice daily
- At the high end of the dosing range, tolerability may be enhanced by splitting IR dose into 3 or more divided doses

- Extended-release formulation of pregabalin allows for once daily dosing
- Extended-release tablet should be swallowed whole and should not be split, crushed, or chewed
- For intolerable sedation with the immediate-release formulation, can give most of the dose at night and less during the day or switch to extended-release
- To improve slow-wave sleep, may only need to take pregabalin at bedtime
- May be taken with or without food

### **Overdose**

- No fatalities

### **Long-Term Use**

- Safe

### **Habit Forming**

- No

### **How to Stop**

- Taper over a minimum of 1 week

- Epilepsy patients may seize upon withdrawal, especially if withdrawal is abrupt
- Discontinuation symptoms uncommon

### **Pharmacokinetics**

- Pregabalin is not metabolized but excreted intact renally
- Elimination half-life approximately 5–7 hours

### **Drug Interactions**

- Pregabalin has not been shown to have significant pharmacokinetic drug interactions
- Because pregabalin is excreted unchanged, it is unlikely to have significant pharmacokinetic drug interactions
- May add to or potentiate the sedative effects of oxycodone, lorazepam, and alcohol

### **Other Warnings/Precautions**

- Depressive effects, including respiratory depression, may be increased by other CNS depressants (opioids, benzodiazepines, alcohol, MAOIs, other anticonvulsants, etc.)
- Use lowest possible dose of pregabalin and monitor for symptoms of respiratory depression if patient is taking concomitant CNS depressant, has underlying respiratory disease, or is elderly

- Dizziness and sedation could increase the chances of accidental injury (falls) in the elderly
- Increased incidence of hemangiosarcoma at high doses in mice involves platelet changes and associated endothelial cell proliferation not present in rats or humans; no evidence to suggest an associated risk for humans
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such side effects immediately

### **Do Not Use**

- If there is a proven allergy to pregabalin or gabapentin
- If patient has a problem of galactose intolerance, the Lapp lactase deficiency, or glucose-galactose malabsorption

## **Special Populations**

### **Renal Impairment**

- Pregabalin is renally excreted, so the dose may need to be lowered
- Dosing can be adjusted according to creatinine clearance, such that patients with clearance below 15 mL/min should receive 25–75 mg/day in 1 dose, patients with clearance between 15–29 mL/min should receive 25–150 mg/day in 1–2 doses, and patients with

clearance between 30–59 mL/min should receive 75–300 mg/day in 2–3 doses

- Starting dose should be at the bottom of the range; titrate as usual up to maximum dose
- Can be removed by hemodialysis; patients receiving hemodialysis may require a supplemental dose of pregabalin following hemodialysis (25–100 mg)

### **Hepatic Impairment**

- Dose adjustment not necessary

### **Cardiac Impairment**

- No specific recommendations

### **Elderly**

- Some patients may tolerate lower doses better
- Elderly patients may be more susceptible to adverse effects



### **Children and Adolescents**

- Approved for partial onset seizures in children ages 1 month and older
- Safety and efficacy have not been established in mental illness indications

- Use should be reserved for the expert

### **Pregnancy**

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- In animal studies, developmental toxicity (including fetal structural abnormalities, skeletal malformations, retarded ossification, and decreased fetal body weight) was observed when pregabalin was administered to pregnant animals at doses greater than or equal to 16 times the maximum recommended human dose (MRHD)
- Use in women of childbearing potential requires weighing potential benefits to the mother against the risks to the fetus
- Antiepileptic Drug Pregnancy Registry: 1-888-233-2334, [www.aedpregnancyregistry.org](http://www.aedpregnancyregistry.org)
- Taper drug if discontinuing
- Seizures, even mild seizures, may cause harm to the embryo/fetus

### **Breast Feeding**

- Unknown if pregabalin is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- \* Recommended either to discontinue drug or bottle feed
- If drug is continued while breast feeding, infant should be monitored for possible adverse effects
- If infant becomes irritable or sedated, breast feeding or drug may need to be discontinued

## **The Art of Psychopharmacology**

### **Potential Advantages**

- First-line for diabetic peripheral neuropathy
- Fibromyalgia
- Anxiety disorders
- Sleep
- Has relatively mild side effect profile
- Has few pharmacokinetic drug interactions
- More potent and probably better tolerated than gabapentin

### **Potential Disadvantages**

- Requires 2–3 times a day dosing
- Not approved for anxiety disorders in the USA

- Not approved for fibromyalgia in Europe

### **Primary Target Symptoms**

- Seizures
- Pain
- Anxiety

### **Pearls**

- \* First treatment approved for fibromyalgia
- \* One of the first treatments approved for neuropathic pain associated with diabetic peripheral neuropathy
- Also approved in postherpetic neuralgia
- Improves sleep disruption as well as pain in patients with painful diabetic peripheral neuropathy or postherpetic neuralgia
- Improves sleep disruption as well as pain associated with fibromyalgia
- Well studied in epilepsy, peripheral neuropathic pain, and GAD, and actually approved for GAD in Europe
- \* Off-label use for GAD, panic disorder, and social anxiety disorder may be justified in the USA
- May have uniquely robust therapeutic actions for both the somatic and the psychic symptoms of GAD
- \* Off-label use as an adjunct for bipolar disorder may not be justified



- \* One of the few agents that enhances slow-wave delta sleep, which may be helpful in chronic neuropathic pain syndromes
- Pregabalin is generally well tolerated, with only mild adverse effects
- \* Although no head-to-head studies, appears to be better tolerated and more consistently efficacious at high doses than gabapentin
- \* Drug absorption and clinical efficacy may be more consistent at high doses for pregabalin compared to gabapentin because of the higher potency of pregabalin and the fact that, unlike gabapentin, it is transported by more than one transport system

### *Suggested Reading*

Lauria-Horner BA , Pohl RB . Pregabalin: a new anxiolytic . *Expert Opin Investig Drugs* 2003 ;**12** :663–72 .

Moore RA , Straube S , Wiffen PJ , Derry S , McQuay HJ . Pregabalin for acute and chronic pain in adults . *Cochrane Database Syst Rev* 2009 ;**8** (3 ):CD007076 .

Stahl SM . Anticonvulsants and the relief of chronic pain: pregabalin and gabapentin as alpha(2)delta ligands at voltage-gated calcium channels . *J Clin Psychiatry* 2004 ;**65** :596–7 .

Stahl SM . Anticonvulsants as anxiolytics, part 2: pregabalin and gabapentin as alpha(2)delta ligands at voltage-gated calcium channels . *J Clin Psychiatry* 2004 ;**65** :460–1 .